Abioye Jesse Fredrick

Java Backend Developer · Distributed Systems · Security Engineering

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PROFILE

Java backend developer with hands-on experience building secure, distributed systems using Spring Boot, RabbitMQ, Redis, and MariaDB. Holds a B.Sc. in Pure Mathematics — formal reasoning applied directly to software design. Projects cover full-stack e-commerce deployment, structural tenant isolation, JWT authentication with token reuse detection, event-driven pipelines with failure handling, and a hand-coded symbolic math engine deployed on Railway. Every architectural decision is documented — what was chosen, what was rejected, and why.

PROJECTS

BarnCart · Spring Boot 3.5 · React · MySQL · Stripe · WebSocket · Docker · 2026

Farm-to-customer e-commerce platform built and deployed end to end — auth, live inventory, Stripe checkout, delivery scheduling, disputes, and an admin panel. Frontend and backend in separate repos, both deployed live.

- Real-time inventory via WebSocket — stock levels update across all sessions the moment a batch is restocked or purchased
- Stripe checkout with a timed reservation — slot held for 15 minutes; expired reservations release stock automatically
- Multi-item checkout bug traced to Hibernate's clearAutomatically wiping unflushed inserts mid-loop — fixed via snapshot pattern and saveAndFlush(), then same assumption traced and corrected across six other call sites
- Admin panel: order fulfillment, dispute management, batch restock, analytics dashboard, delivery slot configuration
- 15 ADRs + 21 bugs documented with root cause, fix applied, and consequence recorded
- Deployed: React frontend on Vercel, Spring Boot backend on Render, Aiven-managed MySQL 8.4

Multi-Tenant Secure API · Java 21 · Spring Boot 3.5 · MariaDB · Redis · JWT · Testcontainers · GitHub Actions · 2026

Production-grade multi-tenant REST API built contract-first. Tenant isolation enforced structurally at the ORM layer — not by convention. A single missed WHERE clause cannot leak data because there is no path to the database that bypasses the tenant scope predicate.

- 9-layer security stack: rate limiting, JWT validation, revocation checking, tenant context, Spring Security, permission checks, repository scoping, ORM immutability, soft delete
- BaseRepository enforces tenant scope on every query — isolation is structural, not a coding convention
- JWT revocation via Redis JTI blacklist on logout; refresh tokens single-use rotating with reuse detection; fails open to DB on Redis outage
- Pure domain workflow engine — TaskWorkflowEngine has zero Spring dependencies, fully unit tested in isolation
- Immutable audit log — state transition and audit entry written in the same transaction; all fields updatable = false
- 39 automated tests across unit, integration, and full HTTP stack against real MariaDB and Redis via Testcontainers
- GitHub Actions CI — all 39 tests run on every push to main
- 10 ADRs + 16 bugs documented with root cause and fix applied

Trust-Aware Incident Intelligence API · Java 23 · Spring Boot 3.5 · MariaDB · Redis · RabbitMQ · JWT · 2026

REST API where every request is evaluated by both role and behavioural trust score. The architectural core — role and trust score never influence each other — is enforced as a hard constraint, not a convention.

- JWT auth with 5-minute access token TTL and opaque refresh tokens stored SHA-256 hashed in the database; token reuse detection revokes the entire token family atomically and returns 401 TOKEN_REUSE_DETECTED
- Redis rate limiting across four trust tiers (TRUSTED / STANDARD / RESTRICTED / BLOCKED); fail-open by design — auth has zero Redis dependency
- Transactional outbox pattern ensures incident creation events reach RabbitMQ even on broker failure — event and outbox row written in one transaction

- Correlation ID generated at servlet level traces every request across filters, services, error responses, and audit logs
- TrustScoreService is the single mutation point for all score changes; every mutation writes to trust_score_history and audit_log in the same transaction
- Flyway owns the schema; Hibernate DDL is disabled; V1–V8 migrations documented

Async Incident Processing Pipeline · Java 21 · Spring Boot 3.5 · RabbitMQ · MariaDB · Docker · 2026

Event-driven consumer that processes incident events published by the Trust API. Six failure scenarios are explicitly handled, each with a documented recovery path verified by database query.

- Retry logic: exponential backoff at 2s / 4s / 8s; messages exceeding 3 attempts are routed to DLQ and marked DEAD — strategy documented in ADR-002
- ProcessingWatchdog polls for records stuck in PROCESSING beyond 8 seconds and transitions them to PENDING_RETRY
- Idempotency enforced via eventId uniqueness — duplicate events are acknowledged and skipped without processing
- RabbitMQ broker failure handled via outbox poller with publisher confirms; no data loss on restart
- Database unavailability caught as DataAccessException and requeued — not routed to DLQ
- ADRs written for broker choice, retry strategy, idempotency approach, and outbox integration

Calculus API · Java 21 · Spring Boot 3.4 · HTML · CSS · JavaScript · Railway · 2026

REST API exposing a symbolic mathematics engine written entirely from scratch — no external math libraries. Spring is a thin HTTP wrapper; the engine has zero framework dependencies.

- Hand-coded tokenizer, recursive descent parser, AST, differentiator, integrator, simplifier, and printer
- Supports differentiation, integration by parts, implicit differentiation, variable exponents, and inverse trig functions
- Engine is testable in isolation without starting the Spring context
- Live frontend deployed publicly at calculus-api-production.up.railway.app

Student Management System · Java 17 · Swing · JDBC · MySQL — No Framework · 2025

Desktop application managing the full academic lifecycle across a multi-admin institution. Built without Spring, Hibernate, or any ORM — every dependency wired manually to understand what frameworks abstract away.

- Four strict layers: UI → Service → DAO → Database; SwingWorker threading enforced across 20 frames — the EDT never blocks
- Multi-admin data isolation enforced at the SQL level on every query via adminId sentinel pattern — eliminates an entire class of data leakage bugs
- Full code review: 15 critical and 8 high-severity bugs identified, root-caused, and fixed; every issue documented with cause and fix

TECHNICAL SKILLS

LanguagesJava 23, Java 21, Java 17, SQL, JavaScript
FrameworksSpring Boot 3.5, Spring Security, Spring Data JPA
DatabasesMariaDB, MySQL — schema design, Flyway migrations, index-aware query design
MessagingRabbitMQ — exchanges, DLQ, retry, publisher confirms, transactional outbox
CachingRedis — rate limiting, TTL, key schema, fail-open design
Auth / SecurityJWT (stateless), opaque refresh tokens, token reuse detection, role-based access
TestingJUnit 5, Testcontainers — unit, integration, full HTTP stack
ObservabilitySpring Actuator, structured JSON logging (Logstash), correlation ID tracing
InfrastructureDocker, Docker Compose, GitHub Actions CI, Railway, Render, Vercel
ToolsMaven, Flyway, Lombok, Git
ConceptsDistributed systems, event-driven architecture, idempotency, failure semantics, ADRs
FrontendHTML, CSS, JavaScript (vanilla) — live frontend on Calculus API; React (BarnCart)

EXPERIENCE

Java Instructor · NIIT Institute · 2025

- Delivered Java programming instruction at certificate level to beginner learners
- Structured complex technical concepts into accessible, sequential curriculum — the same skill applied to project documentation and ADRs

EDUCATION

B.Sc. (Ed) Pure Mathematics · Completed · 2025

Core areas: abstract algebra, linear algebra, real and complex analysis, functional analysis, topology, ODE and PDE, probability, statistics, and applied mathematics. Built a symbolic calculus engine in Java as a direct application of analysis — hand-coded differentiator, integrator, and AST with no external libraries.

Diploma in Java Development · NIIT Institute · In Progress

Practical Java development curriculum covering core language, OOP, and application development.

All projects publicly available at github.com/Kunzoick — READMEs document every architectural decision.